

VANSHIKA RAI

✉ vanshikarai4040@gmail.com

in [vanshika-rai-2b984a338/](https://www.linkedin.com/in/vanshika-rai-2b984a338/)

🌐 [Vanshikarai29](https://github.com/Vanshikarai29)

SKILLS

Languages: Python, SQL, Java

Machine Learning: Supervised & Unsupervised Learning, Regression, Classification, Feature Engineering, Model Evaluation, Cross-Validation, Hyperparameter Tuning

Deep Learning & CV: CNNs, Neural Networks, Transfer Learning, OpenCV, Image Preprocessing

Libraries: NumPy, Pandas, Scikit-learn, TensorFlow, Keras

Data Analysis: EDA, Statistical Analysis, Matplotlib, Seaborn

ML Deployment: REST APIs (Flask/FastAPI), Model Serialization (Pickle, Joblib)

Tools: Git, GitHub, Jupyter Notebook, VS Code

Math for ML: Linear Algebra, Probability, Statistics

PROJECTS

Facial Emotion Detection System

Nov 2025

Machine Learning & Computer Vision

Python, OpenCV, CNN

- Built a **CNN-based real-time facial emotion recognition system** achieving **92% accuracy** across 5 emotion classes.
- Optimized inference pipeline, reducing latency by **30%** for real-time video processing.
- **GitHub:** github.com/Vanshikarai29/face-expression-detector

Real Estate Price Prediction System

Aug 2025

Machine Learning / Data Science

Python, Scikit-learn, Pandas

- Developed and compared regression models, achieving **15% RMSE reduction** through feature engineering.
- Improved model generalization using **cross-validation and hyperparameter tuning**.
- **GitHub:** github.com/Vanshikarai29/real-estate-price-prediction

Smart Placement Preparation Assistant

Jul 2025

AI/ML, Full Stack Development

React, Tailwind CSS, Firebase, Python

- Designed an **AI-driven platform** delivering personalized study plans, interview resources, and company-specific recommendations to optimize placement preparation.
- Implemented interactive frontend and backend workflows using **React, Tailwind CSS, and Firebase**, integrating AI logic for dynamic, company-specific resource delivery.

EDUCATION

ABES Engineering College

2023 – 2027

Bachelor of Technology (B.Tech), Computer Science - CGPA - 8.2/10 (Present)

Ghaziabad, India

Training

AI & Design Thinking Workshop

2025

MeitY, Government of India

Completed a **3-week government program** on **AI fundamentals, supervised learning, and human-centered design**.

Developed **mini AI prototypes** using **data preprocessing, model training, and evaluation** within the **design thinking lifecycle**.

CERTIFICATIONS

- IBM Artificial Intelligence Fundamentals Certification — IBM SkillsBuild
- AI and Design Thinking Workshop — MeitY, Government of India
- Achieved AIR 3789 in NCAT (Top 4% Nationally, 1L+ Participants)
- GUVI-HCL Hackathon Participant (Organized by AKTU)
- ACPC 2024 Competitive Programming Contest — ABES Engineering College
- Git and GitHub Proficiency Development Program